

Task 2.2 – Noise Filtering Documentation

Abstract

This task focuses on restoring images corrupted with high-frequency salt noise. The objective is to remove random white noise pixels while preserving the structural features such as edges and line drawings. Median filtering combined with morphological operations was used to effectively remove impulse noise while maintaining image features.

Introduction

Digital images often contain noise due to sensor errors or transmission disturbances. The provided images contain salt noise, appearing as random white pixels. The challenge is to remove these noise pixels without damaging the underlying structures of the image.

Problem Statement

Two corrupted images were provided: `iron_man_noisy.jpg` and `scenary_noise.jpg`. The task was to remove salt noise while preserving edges and structural details.

Methodology

- Load the noisy image using OpenCV.
- Convert the image to grayscale.
- Apply Median Filtering to remove salt noise.
- Use Morphological Opening to remove remaining noise.
- Save the cleaned image.

Results

The resulting images show a significant reduction in salt noise while maintaining the edges and line structures of the drawings.

Conclusion

Median filtering combined with morphological operations provides an effective solution for removing salt noise while preserving important image features.